Data Driven Approach for Detection and Root Cause Identification of Flare Events

Executive Summary

This project aims to develop an adaptive, data-driven framework for the early detection of flare events and identification of their root causes in petroleum refinery processes using multivariate statistical process monitoring techniques. Flare events are critical operational incidents that result in product loss, increased greenhouse gas emissions, reduced energy efficiency, and potential safety hazards. Therefore, early detection of flare events, together with accurate identification of the contributing process variables, is essential for improving refinery safety and operational performance.

The proposed methodology utilizes historical refinery data provided by Hindustan Petroleum Corporation Limited (HPCL), consisting of minute-wise measurements collected from 16 refinery process units, along with flare measurements, controller data, and pressure safety valve (PSV) information. Four progressively advanced monitoring frameworks have been developed and evaluated. The first approach employs a conventional Principal Component Analysis (PCA)-based monitoring framework as the baseline model. The second approach introduces a hierarchical PCA architecture that combines unit-level monitoring models into a plant-wide monitoring framework. The third approach incorporates a moving-window strategy to continuously update the hierarchical PCA models, thereby enabling adaptive monitoring under changing operating conditions. Finally, the proposed framework integrates Hierarchical PCA with Slow Feature Analysis (SFA) within a moving-window framework to capture both static and dynamic process characteristics for improved flare event detection and root cause identification.

The project comprises the following major components:

- **Historian Data Handling:** Development of a Graphical User Interface (GUI) for acquiring and managing refinery historian data. The GUI enables users to upload process data, automatically segregate measurements into unit-wise datasets, and manage process tags before monitoring.
- **Data Pre-processing:** Historical process data collected from the refinery historian are preprocessed by removing invalid measurements, handling missing values, eliminating redundant variables, and standardizing the data before model development.
- **Hierarchical Process Monitoring:** Separate monitoring models are developed for each refinery unit, and the extracted latent features are integrated to construct plant-wide monitoring models capable of capturing both local unit behaviour and inter-unit interactions.
- **Adaptive Moving-Window Monitoring:** A moving-window framework is implemented to continuously update the monitoring models using the most recent process observations,

thereby accommodating non-stationary operating conditions commonly encountered in refinery processes.

- **Dynamic Feature Extraction:** Slow Feature Analysis (SFA) is integrated with hierarchical PCA to separate slowly varying process dynamics from rapidly varying process changes, enabling improved monitoring of dynamic refinery operations.
- **Flare Event Detection:** Flare events are identified using an event-based detection strategy based on the computed flare flow rate. A flare event is declared when the flare flow rate exceeds the specified threshold for a predefined duration. The monitoring framework is considered successful if the monitoring statistic generates at least one alarm within the specified prediction horizon preceding the flare event.
- **Root Cause Identification:** Reconstruction-Based Contribution (RBC) analysis is performed for successfully detected flare events to quantify the contribution of individual process variables. For the proposed Hierarchical PCA–SFA framework, RBC analysis is carried out separately using the slow-feature subspace, fast-feature subspace, and the combined feature space to improve diagnosis accuracy.
- **Performance Evaluation:** The performance of the four monitoring approaches is evaluated and compared in terms of flare event detection capability and root cause identification performance, demonstrating the advantages of hierarchical monitoring, adaptive model updating, and dynamic feature extraction for refinery process monitoring.

Project Deliverables

The complete monitoring framework has been implemented in Python. The developed framework provides an integrated platform for adaptive flare event monitoring and diagnosis and can be extended for real-time deployment in industrial refinery environments.

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1 Introduction

Petroleum refineries are among the most complex industrial systems, comprising multiple interconnected process units operating under stringent safety and environmental regulations. During normal operation, excess hydrocarbons generated due to process disturbances, equipment malfunctions, emergency shutdowns, or pressure relief events are directed to the flare system to ensure safe disposal. Although flaring is an essential safety mechanism, frequent or excessive flaring results in significant economic losses, increased greenhouse gas emissions, reduced energy efficiency, and environmental pollution. Consequently, minimizing flare events and identifying their underlying causes have become critical objectives for modern refinery operation.

Flare events are generally preceded by subtle changes in process variables distributed across multiple refinery units. Owing to the large number of sensors, complex interactions among process units, and continuously varying operating conditions, identifying the root cause of a flare event is a challenging task for plant operators. Traditional alarm systems are primarily threshold-based and often generate multiple alarms simultaneously, making it difficult to determine the initiating process disturbance. Therefore, there is a growing need for intelligent monitoring frameworks capable of providing early warning of flare events while accurately identifying the process variables responsible for abnormal operating conditions.

Multivariate statistical process monitoring (MSPM) techniques such as Principal Component Analysis (PCA) have been widely employed for fault detection in industrial processes due to their ability to extract latent information from highly correlated process variables. Hotelling's T^2 statistic computed in the reduced-dimensional feature space enables effective detection of abnormal operating conditions, while Reconstruction-Based Contribution (RBC) analysis facilitates root cause identification by quantifying the contribution of individual process variables. However, conventional PCA-based monitoring considers the entire refinery as a single process and does not explicitly exploit the hierarchical structure of large-scale industrial plants. Furthermore, static PCA models are unable to adequately capture the non-stationary behaviour commonly observed in refinery operations.

To address these limitations, hierarchical monitoring strategies have recently gained considerable attention. By constructing separate monitoring models for individual process units and subsequently integrating their latent features into a plant-wide monitoring framework, hierarchical approaches preserve local process characteristics while simultaneously capturing interactions among different refinery units. Nevertheless, batch-mode hierarchical models remain unsuitable for continuously changing operating conditions because the monitoring models are developed offline and are not updated as new process data become available.

Adaptive monitoring techniques based on moving-window modelling provide an effective solution to this problem. By continuously updating the monitoring model using the most recent observations, moving-window approaches can effectively track gradual process changes while maintaining reliable monitoring performance. Furthermore, Slow Feature Analysis (SFA) provides an attractive framework for dynamic process monitoring by extracting slowly varying latent features that represent the dominant process dynamics and separating them from rapidly varying components associated with process disturbances. Integrating SFA with hierarchical PCA enables improved monitoring of nonlinear and time-varying refinery processes while enhancing the capability for early flare event detection.

1.1 Objectives of the Project

The primary objective of this project is to develop an adaptive data-driven framework for early flare event detection and root cause identification in petroleum refinery processes using multivariate statistical process monitoring techniques.

The specific objectives are as follows:

1. To develop a graphical user interface (GUI) for efficient acquisition and management of historian data from the refinery database.
2. To develop a baseline PCA-based monitoring framework for flare event detection and root cause identification.
3. To design a hierarchical PCA framework that preserves unit-level information while monitoring plant-wide operating conditions.
4. To incorporate a moving-window strategy for adaptive updating of the monitoring models under non-stationary refinery operating conditions.
5. To integrate Slow Feature Analysis (SFA) with hierarchical PCA for improved dynamic process monitoring and early flare event detection.
6. To compare the performance of the proposed monitoring approaches in terms of flare event detection capability and root cause identification accuracy.

Organization of the Report

The remainder of this report is organized as follows. Section 2 presents a comprehensive literature survey on existing approaches for flare event detection and root cause identification. Section 3 introduces the preliminary concepts required for this work, including Principal Component Analysis (PCA), Slow Feature Analysis (SFA), Hotelling's T^2 statistic, Reconstruction-Based Contribution (RBC) analysis, the moving-window strategy, and the flare event detection framework. Section 4 describes the proposed methodology, including the historian data handling framework, data preprocessing, monitoring strategy, and the four progressively developed monitoring approaches. Section 5 presents the implementation details, experimental results, and performance evaluation of the proposed monitoring frameworks using historical refinery data, followed by a comparative analysis of the developed approaches. Finally, Sections 6 and 7 summarize the major findings of the project, present the conclusions, and discuss potential directions for future research.

2 Literature Survey

Gas flaring is an indispensable safety mechanism employed in petroleum refineries and petrochemical industries to relieve excess pressure generated during abnormal operating conditions. Although flaring prevents catastrophic failures, it results in substantial greenhouse gas emissions, energy losses, and environmental pollution. Consequently, significant research efforts have focused on the early detection, prediction, and monitoring of flare events using data-driven, machine learning, and computer vision techniques. The summarized literature survey is presented in Table 1.

Fang et al. [2] proposed one of the earliest data-driven frameworks for early prediction of gas flare events using a hierarchically distributed monitoring architecture. Their work was motivated by the limitations of conventional centralized Principal Component Analysis (PCA), which could identify only a limited number of flare events. The proposed methodology divided the refinery into multiple process units and extracted local dynamic information using Principal Component Analysis (PCA), Slow Feature Analysis (SFA), and Wavelet Transform (WT). These local monitoring statistics were subsequently integrated within a hierarchical monitoring framework to provide plant-wide flare prediction. The methodology was validated using one year of industrial refinery data containing 132 process variables from 18 operating units and successfully predicted the majority of flare events while maintaining an acceptable false alarm rate. However, the framework relied mainly on linear latent variable models and did not explicitly account for nonlinear process dynamics or adaptive changes in operating conditions.

To address these limitations, Memarian et al. [6] introduced a unified unit-wise and plantwide monitoring framework for early flare event detection. Unlike previous studies that focused solely on distributed monitoring, the proposed approach simultaneously captured local unit dynamics and plant-wide nonlinear interactions. At the unit level, the authors employed Moving Window Weighted Probabilistic Slow Feature Analysis (MWWPSFA), enabling adaptive monitoring under time-varying operating conditions. Plant-wide nonlinear dynamics were modeled using a Variational Autoencoder integrated with Slow Feature Analysis (VAE-SFA), which extracted nonlinear latent features from Markov Transition Field representations of process variables. The framework demonstrated superior detection accuracy and robustness on industrial refinery data by effectively capturing slowly varying faults and process drift. Nevertheless, the proposed deep learning architecture increased computational complexity and required extensive offline model training, potentially limiting real-time deployment in large-scale industrial environments.

While process-data-based monitoring has gained considerable attention, vision-based approaches have emerged as promising alternatives for directly monitoring flare combustion characteristics. Gu et al. [4] proposed a Vision-Based Monitor of Flare Soot (VMFS) for automatic detection of soot emissions using image processing techniques. Instead of relying on deep neural networks, the proposed system combined broadly tuned color channel analysis, saliency detection, K-means clustering, and background color analysis to accurately identify flame and soot regions from industrial video streams. Experimental validation on real petrochemical plant videos demonstrated that VMFS achieved higher monitoring accuracy and computational efficiency than existing image-processing methods. The authors argued that conventional deep learning methods were unsuitable because of limited abnormal image samples and poor robustness to image distortions. However, the proposed approach focused exclusively on soot monitoring and could not provide early prediction of flare events or identify their underlying process causes.

Recent advances in machine learning have also enabled predictive modeling of flare combustion performance. Pandiyan et al. [7] developed comprehensive supervised machine learning models to predict combustion efficiency (CE) and destruction efficiency (DE) for steam-assisted, air-assisted, and non-assisted industrial flares. The study compiled a database comprising over 1900 operating conditions collected from diverse experimental studies and evaluated six supervised learning algorithms. Bayesian optimization was employed for hyperparameter tuning, while physics-informed features were incorporated to improve prediction accuracy. Extreme Gradient Boosting (XGBoost) consistently outperformed alternative models, achieving testing coefficients of determination exceeding 0.93 for steam-assisted flares and 0.97 for air-assisted flares. The work demonstrated that machine learning provides an economical alternative to expensive large-scale flare experiments. Nevertheless, the study primarily focused on combustion performance prediction rather than real-time flare event detection or process monitoring, and prediction accuracy

for non-assisted flares remained comparatively lower because of missing operational features.

Extending vision-based monitoring further, Boumaraf et al. [1] proposed an automatic air-flow monitoring system for industrial flare stacks using Deep Convolutional Neural Networks (DCNNs). A laboratory-scale flare system was designed to generate images corresponding to different air-flow conditions, and twelve pre-trained deep learning models were evaluated using transfer learning. Among them, EfficientNetB7 combined with Adaptive Histogram Equalization (AHE) achieved the highest classification accuracy of 99.04% and an F1-score of 98.85% for recognizing closed, half-opened, and fully-opened air-flow conditions. The proposed framework demonstrated the feasibility of replacing expensive physical air-flow sensors with computer vision systems and highlighted its potential integration with AI, IoT, and Digital Twin technologies for real-time flare optimization. However, the study was conducted using laboratory-scale data and addressed only air-flow classification without considering plant-wide process monitoring or early flare event prediction.

Overall, the reviewed literature demonstrates a clear evolution from conventional multivariate statistical monitoring techniques toward advanced machine learning and deep learning approaches. Early research primarily focused on process-variable-based flare event prediction using distributed latent variable models, whereas recent studies increasingly incorporate nonlinear representation learning, computer vision, and artificial intelligence to improve flare monitoring and operational efficiency. Nevertheless, most existing approaches address either process monitoring, combustion performance prediction, or visual monitoring independently. A unified framework capable of simultaneously performing early flare event detection, identifying root causes, and providing interpretable diagnostics for large-scale refinery operations remains an important research challenge.

Table 1: Comparative Analysis of Existing Flare Monitoring and Prediction Methods

Ref.	**Objective**	**Methodology**	**Dataset**	**Key Findings**	**Limitations**
Fang et al. (2019) [2]	Early prediction of gas flare events	Hierarchical distributed monitoring using PCA, SFA and Wavelet Transform	Real refinery data (132 variables, 18 units)	First plant-wide framework for early flare prediction; successfully predicted most flare events with acceptable false alarms.	Linear latent-variable model; limited handling of nonlinear and time-varying process dynamics.
Memarian et al. (2024) [6]	Early detection of gas flare events	Moving Window Weighted PSFA (MWWPSFA) and VAE-SFA for unit-wise and plantwide monitoring	Industrial refinery data	Adaptive monitoring framework captures local dynamics and plantwide nonlinear behaviour, improving detection accuracy and robustness.	Higher computational complexity and offline training requirement for deep learning model.
Gu et al. (2020) [4]	Automatic flare soot monitoring	Vision-based image processing using flame localization, saliency detection and K-means clustering	Industrial flare video sequences	Accurate and computationally efficient soot detection without requiring deep neural networks.	Limited to soot monitoring; does not predict flare events or identify process faults.
Pandiyan et al. (2025) [7]	Prediction of flare combustion performance	Supervised ML (XGBoost, RF, ANN, etc.) with Bayesian optimization	1900+ flare operating conditions	XGBoost achieved high prediction accuracy ($R^2 > 0.93$) for combustion and destruction efficiency.	Focuses on combustion performance rather than real-time flare event prediction.
Boumaraf et al. (2025) [1]	Vision-based air-flow monitoring	Transfer learning using twelve DCNN models (EfficientNetB7 best)	Laboratory flare image dataset	Achieved 99.04% classification accuracy for air-flow levels; suitable for AI and Digital Twin applications.	Validated only on laboratory data; does not perform plant-wide process monitoring or flare prediction.

3 Preliminaries

3.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a powerful unsupervised learning technique used to reduce the dimensionality of large and complex datasets [9]. The primary goal of PCA is to discover a lower-dimensional space that captures the essential structure and patterns in the data without losing significant information. It does this by identifying the directions in which the data varies the most known as principal components and transforming the original data into a new coordinate system aligned with these directions. To discuss the concept of PCA, let's consider the data set which is in the form of a data matrix:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_N \end{bmatrix} \tag{1}$$

where $\mathbf{x} = \mathbf{x}_j \in \mathbb{R}^n$, $j = 1, .., N$ are the observations at N time instants. Each observation consists of the measurements of n process variables. We can also represent the observations as row vectors which results in the data matrix:

$$\mathbf{X}^T = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{bmatrix} \tag{2}$$

where $\mathbf{x}_j^T$, $j = 1, 2, ..., N$ are row vectors of dimension n. In the rest of the report, we represent the data in the form of Eq. 1 with observations as column vectors. We also assume the data is preprocessed. Here by preprocessing, we mean cleaning and normalizing the raw data to make it suitable for the PCA algorithm. By normalization, we mean shifting and scaling the observations appropriately to use a common scale for the data set. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be an observation with n PVs. Then we can obtain the normalized observation as

$$\mathbf{x}_{normalized} = \begin{bmatrix} \frac{x_1 - E(x_1)}{\sigma_{x_1}} \\ \frac{x_2 - E(x_2)}{\sigma_{x_2}} \\ \vdots \\ \frac{x_n - E(x_n)}{\sigma_{x_n}} \end{bmatrix} \tag{3}$$

where $E(x_i)$ and σ_{x_i}, $i = 1, 2, ..., n$ are the mean and standard deviation of the ith PV. We denote the covariance matrix of the data as $\Sigma_{\mathbf{x}}$ which is computed as the sample covariance matrix:

$$\Sigma_{\mathbf{x}} = \frac{1}{N-1}\mathbf{X}\mathbf{X}^T \tag{4}$$

Next we move on to the formulation of the PCA optimization problem and its solution. Let $\mathbf{P} = \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{bmatrix}$ be the orthogonal matrix with the column vectors spans $\mathbb{R}^n$. Then each observation $\mathbf{x} \in \mathbb{R}^n$ can be represented as

$$\mathbf{x} = \mathbf{P}\mathbf{y} = \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \mathbf{p}_1 y_1 + \mathbf{p}_2 y_2 + ... + \mathbf{p}_n y_n. \tag{5}$$

From which the transformed observation $\mathbf{y}$ is obtained as

$$\mathbf{y} = \mathbf{P}^{-1}\mathbf{x} = \mathbf{P}^T\mathbf{x}. \tag{6}$$

In PCA, the objective is to find the optimal basis $\mathbf{P}$ which maximizes the variance of the transformed observations or features. For example, we can consider the 2D case as shown in Figure 3 in which the first principal component $\mathbf{p}_1$ is chosen for maximizing the variance of the transformed feature $y_1 = \mathbf{p}_1^T\mathbf{x}$ and the second principal component $\mathbf{p}_2$ is chosen for maximizing the variance of $y_2 = \mathbf{p}_2^T\mathbf{x}$. In general for $n-$ dimensional case, we have to find the n principal components. Let $\mathbf{P} = \begin{bmatrix}\mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n\end{bmatrix}$ be the matrix with the principal components as its columns which maximizes the variance $\Sigma_{y_1} + \Sigma_{y_2} + ... + \Sigma_{y_n} = Trace\,(\Sigma_{\mathbf{y}})$. From this we can

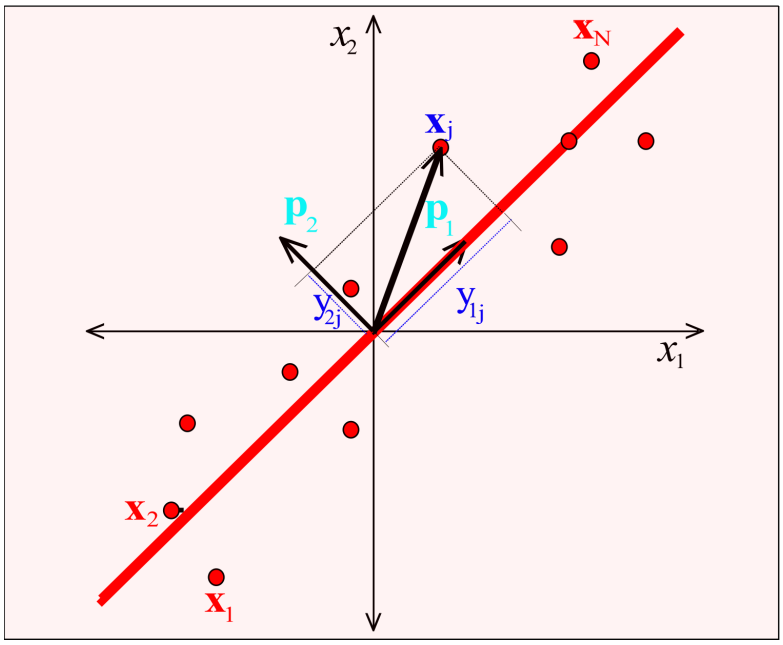


Figure 1: PCA illustration.

formulate the optimization problem for PCA as follows:

$$\begin{aligned} &\max_{\mathbf{P}} \; Trace\,(\Sigma_{\mathbf{y}}) \quad subject\ to \\ &\mathbf{P}^T\mathbf{P} = \mathbf{I}. \end{aligned} \tag{7}$$

Now from Eq. 6 we have

$$\Sigma_{\mathbf{y}} = \Sigma_{\mathbf{P}^T\mathbf{x}} = \mathbf{P}^T\Sigma_{\mathbf{x}}\mathbf{P} \tag{8}$$

which is based on the formula. $\Sigma_{\mathbf{Ax}} = \mathbf{A}\Sigma_{\mathbf{x}}\mathbf{A}^T$ [5]. Substituting (8) in the optimization problem (7) results in:

$$\begin{aligned} &\max_{\mathbf{P}} \; Trace\,(\mathbf{P}^T\Sigma_{\mathbf{x}}\mathbf{P}) \quad subject\ to \\ &\mathbf{P}^T\mathbf{P} = \mathbf{I}. \end{aligned} \tag{9}$$

which is a constrained optimization problem with a quadratic cost function and equality constraints. The constrained optimization problem can be reformulated as an unconstrained optimization problem using the Lagrangian multiplier matrix $\mathbf{\Lambda} \in \mathbb{R}^{n\times n}$:

$$\max_{\mathbf{P}} \; J = Trace\,(\mathbf{P}^T\Sigma_{\mathbf{x}}\mathbf{P}) + Trace\,(\mathbf{\Lambda}[\mathbf{I} - \mathbf{P}^T\mathbf{P}]). \tag{10}$$

It can be shown that $\mathbf{\Lambda}$ is a diagonal matrix for the above optimization problem (See Appendix B.2 of [3] for the proof) and can be represented as $\mathbf{\Lambda} = diag\,(\lambda_1, \lambda_2, ..., \lambda_n)$. Now, the first-order condition for optimality gives $\nabla J_{\mathbf{P}} = 0$ which results in:

$$2\Sigma_{\mathbf{x}}\mathbf{P} - 2\mathbf{\Lambda}\mathbf{P} = 0 \quad \Longrightarrow \quad \Sigma_{\mathbf{x}}\mathbf{P} = \mathbf{\Lambda}\mathbf{P}. \tag{11}$$

This shows the principal components for the n dimensional vector $\mathbf{x}$ will be the eigenvectors of $\Sigma_{\mathbf{x}}$ and can be obtained from the eigenvalue decomposition (EVD) of $\Sigma_{\mathbf{x}}$:

$$\Sigma_{\mathbf{x}} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T. \qquad where\, \mathbf{P} = \begin{bmatrix}\mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n\end{bmatrix},\ \mathbf{\Lambda} = \begin{bmatrix}\lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \lambda_n\end{bmatrix}. \tag{12}$$

By considering the first p principal components in $\mathbf{P}$ as significant and the rest $n-p$ as residual components we can represent the observation $\mathbf{x}$ in the principal space and residual space:

$$\mathbf{x} = \mathbf{P}\mathbf{y} = \begin{bmatrix}\mathbf{p}_1 & \dots & \mathbf{p}_p & \mathbf{p}_{p+1} & \dots & \mathbf{p}_n\end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_p \\ y_{p+1} \\ \vdots \\ y_n \end{bmatrix} = \mathbf{P}_p\mathbf{y}_p + \mathbf{P}_r\mathbf{y}_r = \mathbf{x}_p + \mathbf{x}_r \tag{13}$$

where $\mathbf{x}_p, \mathbf{x}_r$ are the projections of the observation $\mathbf{x}$ in to the principal space and residual space respectively. The principal space and residual space are spanned by the column vectors of $\mathbf{P}_p = \begin{bmatrix}\mathbf{p}_1 & \dots & \mathbf{p}_p\end{bmatrix}$ and $\mathbf{P}_r = \begin{bmatrix}\mathbf{p}_{p+1} & \dots & \mathbf{p}_n\end{bmatrix}$ respectively. Here the principal space can be considered as a low-dimensional subspace in which the given data can be represented with a desired amount of variability. From Eqs. (6) and (13) the transformed observations can be obtained as

$$\mathbf{y} = \begin{bmatrix}\mathbf{y}_p \\ \mathbf{y}_r\end{bmatrix} = \mathbf{P}^{-1}\mathbf{x} = \mathbf{P}^T\mathbf{x} = \begin{bmatrix}\mathbf{P}_p^T \\ \mathbf{P}_r^T\end{bmatrix}\mathbf{x}. \tag{14}$$

Substituting for $\mathbf{y}_p$ and $\mathbf{y}_r$, we can rewrite (13) as

$$\mathbf{x} = \mathbf{P}_p\mathbf{y}_p + \mathbf{P}_r\mathbf{y}_r = \mathbf{P}_p\mathbf{P}_p^T\mathbf{x} + \mathbf{P}_r\mathbf{P}_r^T\mathbf{x} = \mathbf{x}_p + \mathbf{x}_r. \tag{15}$$

From this, the residual component is obtained as

$$\mathbf{x}_r = \mathbf{x} - \mathbf{x}_p = \mathbf{x} - \mathbf{P}_p\mathbf{P}_p^T\mathbf{x} = [\mathbf{I} - \mathbf{P}_p\mathbf{P}_p^T]\mathbf{x}. \tag{16}$$

3.2 Hotelling's T^2 Statistic

Hotelling's T^2 statistic is one of the most widely used multivariate statistical indices for process monitoring, fault detection, and fault diagnosis. It measures the distance of an observation from the centre of the normal operating region while accounting for the correlation among variables. For a random vector $\mathbf{z}$ with mean $E(\mathbf{z})$ and covariance matrix $\mathbf{\Sigma}_{\mathbf{z}}$, the Hotelling's T^2 statistic is defined as

$$T_{\mathbf{z}}^2 = \left[\mathbf{z} - E(\mathbf{z})\right]^T \mathbf{\Sigma}_{\mathbf{z}}^{-1} \left[\mathbf{z} - E(\mathbf{z})\right]. \tag{17}$$

The T^2 statistic is a quadratic form. The level sets of a quadratic form are ellipsoids, which represent contours of equal statistical distance from the process mean. To illustrate this, consider the quadratic function

$$\mathbf{x}^T\mathbf{M}\mathbf{x},$$

where

$$\mathbf{M} = \begin{bmatrix} 0.5 & 0.2 \\ 0.2 & 0.5 \end{bmatrix}.$$

Figure 2(a) shows the surface corresponding to $\mathbf{x}^T\mathbf{M}\mathbf{x}$, while Figure 2(b) shows the level curve defined by $\mathbf{x}^T\mathbf{M}\mathbf{x} = 1$, which is an ellipse. This illustrates that observations having the same T^2 value lie on an ellipsoidal contour in the multivariate space.

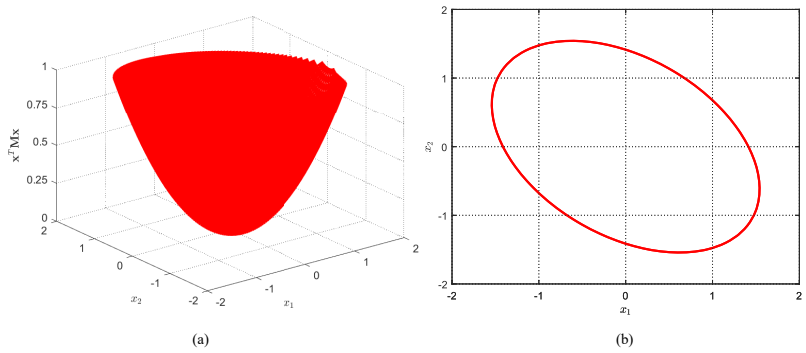


Figure 2: (a) Plot of $\mathbf{x}^T\mathbf{M}\mathbf{x}$. (b) Level curve of $\mathbf{x}^T\mathbf{M}\mathbf{x} = 1$, illustrating the ellipsoidal nature of quadratic forms.

In Principal Component Analysis (PCA), Hotelling's T^2 statistic is computed in the retained principal component subspace. Let $\mathbf{t}$ denote the score vector corresponding to an observation, and let $\mathbf{\Lambda}_r$ be the diagonal matrix containing the eigenvalues of the retained r principal components. The PCA-based Hotelling's T^2 statistic is then given by

$$T^2 = \mathbf{t}^T\mathbf{\Lambda}_r^{-1}\mathbf{t}. \tag{18}$$

This statistic measures the variation of an observation within the principal component subspace retained by the PCA model. Large values of T^2 indicate that the observation lies far from the centre of the normal operating region in the score space.

To determine whether an observation is statistically significant, the computed T^2 value is compared with a confidence limit obtained from the F-distribution. At a significance level α, the confidence limit is

$$T_\alpha^2 = \beta\frac{r(N-1)(N+1)}{N(N-r)}F_\alpha(r, N-r), \tag{19}$$

where N is the number of training samples, r is the number of retained principal components, $F_\alpha(r, N-r)$ denotes the upper α percentile of the F-distribution, and β is an optional tuning parameter used to adjust the sensitivity of the monitoring scheme. An observation is declared abnormal whenever

$$T^2 > T_\alpha^2. \tag{20}$$

3.3 Slow Feature Analysis (SFA)

Slow feature analysis (SFA) [8] is an unsupervised learning algorithm used for extracting the slowly varying features from high-dimensional time series data. SFA can be applied directly to the data, or it can be combined with PCA. In that case, PCA first reduces the complexity of the data, and then SFA is used in the next step to identify the slowly changing features. The data set contains observations that can be represented as the data matrix:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \dots & \mathbf{y}_N \end{bmatrix} \tag{21}$$

where $\mathbf{y} = \mathbf{y}_j \in \mathbb{R}^m, j = 1, .., N$ are the observations at N time instants. Let $\mathbf{Q} = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \dots & \mathbf{q}_m \end{bmatrix}$ with the column vectors spans $\mathbb{R}^m$. Then each $\mathbf{y} \in \mathbb{R}^m$ can be represented as

$$\mathbf{y} = \mathbf{Q}\mathbf{v} = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \dots & \mathbf{q}_m \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{bmatrix} = \mathbf{q}_1 v_1 + \mathbf{q}_2 v_2 + ... + \mathbf{q}_m v_m. \tag{22}$$

From which the transformed observation $\mathbf{v}$ is obtained as

$$\mathbf{v} = \mathbf{Q}^{-1}\mathbf{y} = \mathbf{W}^T\mathbf{y} \tag{23}$$

where $\mathbf{W}^T = \mathbf{Q}^{-1}$ in which we introduced the matrix $\mathbf{W}^T$ to follow the same notations as in PCA derivation. In SFA, the objective is to find the optimal basis $\mathbf{W}^T$ which minimizes the variance of the derivative of the transformed feature $\mathbf{v}$ [10]. This gives the optimization problem for SFA as follows:

$$\begin{aligned} &\min_{\mathbf{W}} \; Trace\left(\Sigma_{\Delta\mathbf{v}}\right) \quad subject\ to \\ &\qquad \Sigma_{\mathbf{v}} = \mathbf{I}. \end{aligned} \tag{24}$$

We have

$$\begin{aligned} \Sigma_{\mathbf{v}} &= \Sigma_{\mathbf{W}^T\mathbf{y}} = \mathbf{W}^T\Sigma_{\mathbf{y}}\mathbf{W} \\ \Sigma_{\Delta\mathbf{v}} &= \Sigma_{\mathbf{W}^T\Delta\mathbf{y}} = \mathbf{W}^T\Sigma_{\Delta\mathbf{y}}\mathbf{W}. \end{aligned} \tag{25}$$

Using this, the optimization problem can be rewritten as:

$$\begin{aligned} &\min_{\mathbf{W}} \; Trace\left(\mathbf{W}^T\Sigma_{\Delta\mathbf{y}}\mathbf{W}\right) \quad subject\ to \\ &\qquad \mathbf{W}^T\Sigma_{\mathbf{y}}\mathbf{W} = \mathbf{I}. \end{aligned} \tag{26}$$

Similar to PCA derivation, using the Lagrangian multipliers $\mathbf{\Omega}$ and first-order condition for optimality gives

$$\Sigma_{\mathbf{y}}\mathbf{W} = \Sigma_{\Delta\mathbf{y}}\mathbf{W}\mathbf{\Omega} \qquad where\ \mathbf{W} = \begin{bmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \dots & \mathbf{w}_m \end{bmatrix},\ \mathbf{\Omega} = \begin{bmatrix} \omega_1 & 0 & \dots & 0 \\ 0 & \omega_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \omega_m \end{bmatrix} \tag{27}$$

which is in the form of a generalized eigenvalue problem and the optimal basis $\mathbf{W}$ is given by the generalized eigenvectors.

By considering the first s eigenvectors as slow components and the rest $m - s$ as fast components we can transform the observation $\mathbf{y}$ into the slow feature space and fast feature space:

$$\mathbf{v} = \begin{bmatrix} \mathbf{v}_s \\ \mathbf{v}_f \end{bmatrix} = \mathbf{Q}^{-1}\mathbf{y} = \mathbf{W}^T\mathbf{y} = \begin{bmatrix} \mathbf{W}_s^T \\ \mathbf{W}_f^T \end{bmatrix} \mathbf{y}. \tag{28}$$

To find the segregation index s which segregates the slow and fast components, we compute the change in eigenvalues $\Delta\omega_i = \omega_{i+1} - \omega_i,\ i = 1, 2, ..., m-1$. We assume the eigenvalues in $\mathbf{\Omega}$ are arranged in increasing order. Therefore $\Delta\omega_i$ gives the rate of increase of the eigenvalues in $\mathbf{\Omega}$. Then the segregation index s is computed as

$$s = arg\ \underset{i}{max}\ \ \Delta\omega_i \tag{29}$$

which gives the index with the largest rate of increase.

3.4 T^2 Statistic for PCA

Transformed observation y_p obtained from PCA,

$$y_p = p_p^T x$$

To standardize (projected data) variance of each principal component direction to one, a transformation is applied

$$z = \Sigma_p^{-1/2} y_p$$

From the equation of T^2

$$\begin{aligned} T_z^2 &= z^T z \\ &= (\Sigma_p^{-1/2} y_p)^T (\Sigma_p^{-1/2} y_p) \\ &= x^T p_p \Sigma_p^{-1} p_p^T x \\ &= \mathbf{y}_p^T \mathbf{\Sigma}_p^{-1} \mathbf{y}_p \end{aligned}$$

3.5 T^2 Statistic for H-PCA SFA

Now consider the slow feature vector $\mathbf{v}_s$ and fast feature $\mathbf{v}_f$ for which the mean and variance are obtained as

$$\begin{aligned} E(\mathbf{v}_s) = \mathbf{W}_s^T E(\mathbf{y}) = 0, \quad \Sigma_{\mathbf{v}_s} = \Sigma_{\mathbf{W}_s^T \mathbf{y}} = \mathbf{W}_s^T \Sigma_y \mathbf{W}_s = \mathbf{I} \\ E(\mathbf{v}_f) = \mathbf{W}_f^T E(\mathbf{y}) = 0, \quad \Sigma_{\mathbf{v}_f} = \Sigma_{\mathbf{W}_f^T \mathbf{y}} = \mathbf{W}_f^T \Sigma_y \mathbf{W}_f = \mathbf{I} \end{aligned} \tag{30}$$

in which we used the relations $E(\mathbf{y}) = 0$ and (28). Using Eqs. (17) and (30) the T^2 statistic for slow and fast feature vectors can be computed as

$$\begin{aligned} T_{\mathbf{v}_s}^2 &= [\mathbf{v}_s - E(\mathbf{v}_s)]^T \Sigma_{\mathbf{v}_s}^{-1} [\mathbf{v}_s - E(\mathbf{v}_s)] = \mathbf{v}_s^T \mathbf{v}_s \\ T_{\mathbf{v}_f}^2 &= [\mathbf{v}_f - E(\mathbf{v}_f)]^T \Sigma_{\mathbf{v}_f}^{-1} [\mathbf{v}_f - E(\mathbf{v}_f)] = \mathbf{v}_f^T \mathbf{v}_f. \end{aligned} \tag{31}$$

3.6 Reconstruction-Based Contribution (RBC)

Reconstruction-based contribution (RBC) is an index that represents the contribution of each variable in terms of the amount of reconstruction required. RBC indicates the amount of reconstruction required for the variable to reach the normal region. For variables with values far from the normal values, the amount of reconstruction required will be more. Consequently, variables with larger RBC values will be the most affected by the fault, i.e. require more reconstruction. Consider the arbitrary data matrix:

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_N \end{bmatrix} \tag{32}$$

with $\mathbf{a} = \mathbf{a}_j \in \mathbb{R}^n$, $j = 1, .., N$ as the observations at N time instants. Here we used a different notation for the data matrix to indicate that the RBC index can be applied to any data in original coordinates or transformed coordinates with PCA, SFA, etc. Let the fault detection index be chosen as a quadratic function:

$$I(\mathbf{a}) = \mathbf{a}^T \mathbf{M} \mathbf{a}. \tag{33}$$

From Eqs (31), it can be observed that most of the fault detection indices can be represented as a quadratic function as in the above equation. We define the reconstructed vector along direction $\mathbf{e}_i, i = 1, .., n$ as

$$\mathbf{z}_i = \mathbf{a} - \mathbf{e}_i r_i \tag{34}$$

where $\mathbf{e}_i$ is the direction of reconstruction and r_i is the amount of reconstruction along the direction r_i. The fault detection index of the reconstructed vector is defined as follows

$$I(\mathbf{z}_i) = \mathbf{z}_i^T \mathbf{M} \mathbf{z}_i. \tag{35}$$

In RBC the objective is to find r_i which minimizes the index $I(\mathbf{z}_i)$. In other words, we are

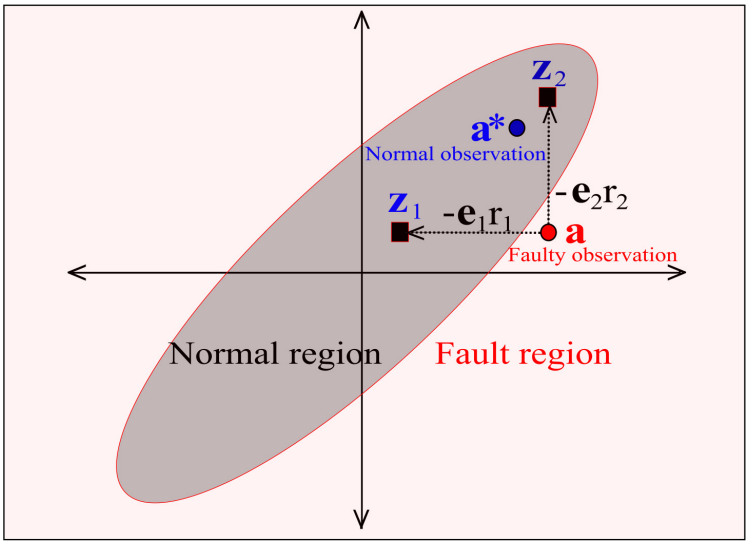


Figure 3: RBC illustration.

trying to find the reconstructed vector in the normal region (i. e., with minimum fault detection index) along the direction $\mathbf{e}_i$. This gives the the optimization problem for RBC:

$$\min_{r_i} \; I(\mathbf{z}_i). \tag{36}$$

We have from Eqs. (34) and (35)

$$I(\mathbf{z}_i) = \mathbf{z}_i^T \mathbf{M} \mathbf{z}_i = [\mathbf{a} - \mathbf{e}_i r_i]^T \mathbf{M} [\mathbf{a} - \mathbf{e}_i r_i] = \mathbf{a}^T \mathbf{M} \mathbf{a} - 2 r_i \mathbf{e}_i^T \mathbf{M} \mathbf{a} + r_r \mathbf{e}_i^T \mathbf{M} \mathbf{e}_i r_i. \tag{37}$$

Using this the optimization problem in Eq. (36) can be rewritten as:

$$\min_{r_i} \ \mathbf{a}^T \mathbf{M} \mathbf{a} - 2 r_i \mathbf{e}_i^T \mathbf{M} \mathbf{a} + r_i \mathbf{e}_i^T \mathbf{M} \mathbf{e}_i r_i. \tag{38}$$

Here the cost function is a quadratic function of the decision variable r_i. Using the first-order condition for optimality gives $\nabla I_{r_i} = 0$ which results in

$$2 r_i \mathbf{e}_i^T \mathbf{M} \mathbf{e}_i - 2 \mathbf{e}_i^T \mathbf{M} \mathbf{a} = 0 \implies r_i = [\mathbf{e}_i^T \mathbf{M} \mathbf{e}_i]^{-1} \mathbf{e}_i^T \mathbf{M} \mathbf{a}. \tag{39}$$

The RBC index is defined as

$$\begin{aligned} RBC_i &= \parallel r_i \mathbf{e}_i \parallel_{\mathbf{M}}^2 = r_i^T \mathbf{e}_i^T \mathbf{M} \mathbf{e}_i r_i \\ &= \mathbf{x}^T \mathbf{M} \mathbf{e}_i [\mathbf{e}_i^T \mathbf{M} \mathbf{e}_i]^{-1} \mathbf{e}_i^T \mathbf{M} \mathbf{e}_i [\mathbf{e}_i^T \mathbf{M} \mathbf{e}_i]^{-1} \mathbf{e}_i^T \mathbf{M} \mathbf{a} = \frac{[\mathbf{e}_i^T \mathbf{M} \mathbf{a}]^2}{\mathbf{e}_i^T \mathbf{M} \mathbf{e}_i}. \end{aligned} \tag{40}$$

3.7 RBC Based on PCA

RBC for T^2

- Calculate $M_s = P_p \cdot Q_y \cdot P_p^T$

  $$M_s = P_p \cdot Q_y \cdot P_p^T$$

  where,

  $$Q_y = \text{cov}(Y^T)^{-1}$$

- Extract the diagonal elements of M_s as a vector M_d

  $$M_d = \text{diag}(M_s)$$

- Calculate $RBCT_k$

  $$\text{RBCT}_k = \frac{(M_s \cdot x_{nk}^T)^2}{M_d}$$

  where, x_{nk} is original normalised data

3.8 RBC Based on HPCA-SFA

- Calculate $M_s = W_s \cdot W_s^T$ using matrix multiplication.

  $$M_s = W_s \cdot W_s^T$$

- Extract the diagonal of M_s to get M_d

  $$M_d = \text{diag}(M_s)$$

- Calculate $RBCT_yk$

  $$\text{RBCT}_{yk} = \frac{(M_s \cdot y_k^T)^2}{M_d}$$

  where,

  $$y_k = x_{nk} \cdot P$$

3.9 Moving Window Implementation

To account for the non-stationary characteristics of industrial processes, a moving window strategy is employed. The adopted approach for moving window is presented in Figure 4.

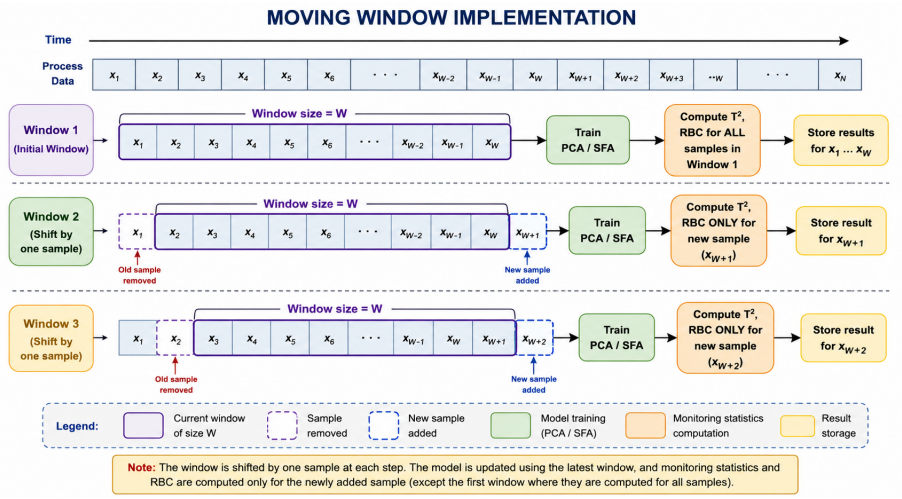


Figure 4: Moving Window strategy

To enable adaptive process monitoring under changing operating conditions, a moving window strategy is adopted, as illustrated in Figure 4. Initially, a PCA (or SFA) model is trained using the first window consisting of the initial W process samples, $\{x_1, x_2, \ldots, x_W\}$. The monitoring statistics, such as Hotelling's T^2 and the Reconstruction-Based Contribution (RBC), are computed for all samples within this initial window and stored. When a new process sample x_{W+1} becomes available, the window is shifted forward by one sample by removing the oldest sample, x_1, and appending the newly acquired sample, x_{W+1}, resulting in the updated window $\{x_2, x_3, \ldots, x_W, x_{W+1}\}$. A new PCA (or SFA) model is then trained using this updated window to reflect the most recent process dynamics. Since the monitoring statistics for the retained samples have already been computed in the previous iteration, only the newly added sample x_{W+1} is evaluated, and its corresponding statistics are stored. In the subsequent iteration, the oldest sample in the current window, x_2, is removed, the next incoming sample, x_{W+2}, is appended, and the window becomes $\{x_3, x_4, \ldots, x_{W+1}, x_{W+2}\}$. This process is repeated sequentially until the final sample x_N is processed. By continuously updating the model using the most recent W samples while computing monitoring statistics only for the newly added observation, the moving window strategy effectively adapts to process variations and significantly reduces the computational burden associated with repeatedly evaluating previously processed samples.

3.10 Flare Event Detection Strategy

To evaluate the effectiveness of the proposed monitoring framework, an event-based flare detection strategy is adopted. The detailed detection is presented in Figure 5.

Let $F(k)$ denote the flare flow rate at sampling instant k, and let F_{th} represent the predefined flare threshold.

A flare event is identified when the flare flow rate exceeds the threshold continuously for at least L consecutive samples, i.e.,

$$F(k) > F_{\text{th}}, \qquad k = k_s, k_s + 1, \ldots, k_s + L - 1, \tag{41}$$

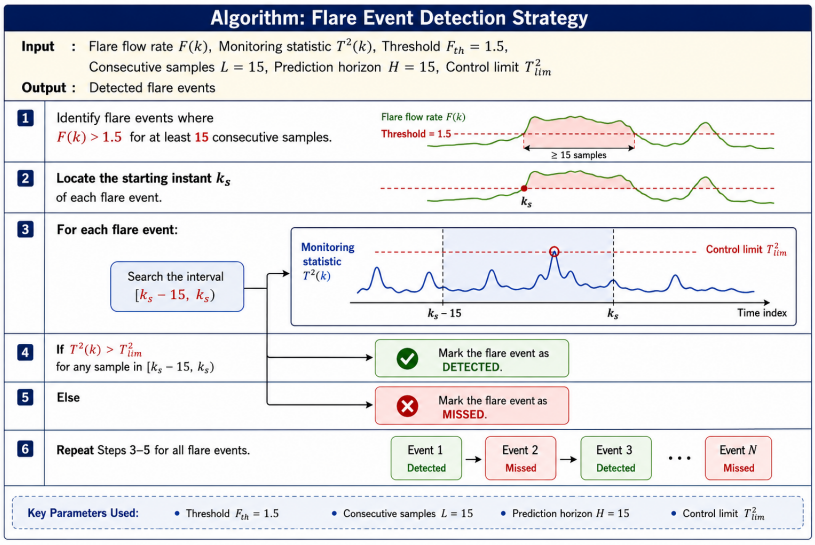


Figure 5: Detection Strategy

where k_s denotes the starting instant of the flare event and L is the minimum duration required to eliminate short-duration fluctuations. In this work, $L = 15$ consecutive samples.

Similarly, let $T^2(k)$ denote the monitoring statistic obtained from the PCA- or SFA-based monitoring model and T^2_{lim} denote its corresponding control limit. A process alarm is generated whenever

$$T^2(k) > T^2_{\text{lim}}. \tag{42}$$

Rather than evaluating individual samples, the monitoring performance is assessed at the event level. For each flare event, the monitoring algorithm is considered to have successfully detected the event if at least one process alarm is generated within a predefined prediction horizon prior to the flare event.

Let H denote the prediction horizon. A flare event beginning at sample k_s is considered successfully detected if there exists at least one sample k satisfying

$$k_s - H \leq k < k_s, \tag{43}$$

such that

$$T^2(k) > T^2_{\text{lim}}. \tag{44}$$

In the present work, a prediction horizon of $H = 15$ minutes is employed. Consequently, a flare event is classified as successfully detected if the monitoring statistic exceeds its control limit at least once during the 15-minute interval preceding the onset of the flare event. This event-based evaluation strategy provides a realistic assessment of early warning capability and avoids multiple detections of the same flare event.

4 Proposed Methodology

This section presents the proposed methodology developed for flare event detection and root cause identification in industrial refinery processes. The proposed framework consists of two major components: (i) a data handling framework for efficient acquisition and management of

process data, and (ii) a monitoring strategy for detecting flare events and identifying their root causes. The monitoring strategy is progressively developed through four approaches, beginning with a conventional PCA-based monitoring framework and subsequently extending it to hierarchical, adaptive moving-window, and hierarchical PCA–SFA-based monitoring frameworks. The overall methodology is designed to provide accurate flare event detection while effectively identifying the process variables responsible for abnormal operating conditions.

4.1 Historian Data

The historical process data used in this work were provided by the HPCL team. The dataset spans the period from 12 April 2025 to 21 May 2026, with measurements recorded at a one-minute sampling interval. The complete dataset consists of 581,774 observations and 376 process variables, providing a comprehensive representation of the refinery operation under various operating conditions.

The process data are organized into 16 refinery units, with each unit containing measurements from pressure, temperature, flow, and other process sensors. The units included in the dataset are listed below:

- **SEU1, SEU2, SEU3:** Solvent Extraction Units
- **DIU:** Diesel Iso-Therming Unit
- **PDA:** Propane De-Asphalting Unit
- **LOUP:** Lube Oil Upgrading Plant
- **PG:** Prime G Unit
- **ISOM:** Isomerization Unit
- **ISOM2:** LPG Mounded Bullets
- **ISOM3:** Propane and VR LPG Treating Unit
- **ISOM4:** Diesel Hydro Treating Unit
- **PDU:** Lube Processing Unit
- **NSU_NHT_CCR:** Continuous Catalyst Regeneration Unit
- **DAK:** De-Aromatized Kerosene Unit
- **CPP:** Captive Power Plant
- **APS_NVPS_LEU (CDU):** Crude Distillation Unit

In addition to the unit-wise process measurements, the historian database contains dedicated datasets for flare system measurements and controller information.

The flare dataset includes the following measurements:

- Fuel Refinery Flare Flow Rate,
- Lube Refinery Flare Flow Rate, and

- Flare Recovery Flow Rate.

Using these measurements, the overall flare flow rate is computed as Flare Recovery Flow Rate.

$$Q_{\text{flare}} = Q_{\text{Fuel}} + Q_{\text{Lube}} - Q_{\text{Recovery}}, \tag{45}$$

where,

- Q_{flare} : Total flare flow rate (MT/h),
- Q_{Fuel}: Fuel refinery flare flow rate (MT/h),
- Q_{Lube}: Lube refinery flare flow rate (MT/h),
- Q_{Recovery} : Flare recovery flow rate (MT/h).

The controller dataset contains the opening percentages of 40 control valves, representing the operating conditions of the major control loops within the refinery. Furthermore, the historian database also includes the Pressure Safety Valve (PSV) set points, which provide important operational information for analysing relief events and identifying potential causes of flare occurrences.

The combination of unit-wise process measurements, flare system data, controller information, and PSV set points provides a comprehensive dataset for the development and evaluation of the proposed flare event detection and root cause identification framework.

4.2 Data Handling Framework

The proposed framework begins with a data handling module that facilitates the acquisition and management of refinery process data. Process measurements are retrieved directly from the plant Distributed Control System (DCS) through a MariaDB database, enabling access to both historical and real-time process data. To simplify data management and improve user interaction, a Graphical User Interface (GUI) has been developed. The GUI enables users to establish a connection with the MariaDB database, select the desired refinery units and monitoring period, and retrieve the corresponding process measurements. In addition, the GUI allows users to include or exclude process tags according to the monitoring requirements. This flexibility enables the removal of redundant, unavailable, or irrelevant process variables before monitoring. The HPCL team provided two folders containing the refinery process data. The first folder contains the process measurements of 16 refinery units, including pressure and temperature sensor data, while the second folder contains the flare and controller data. The developed GUI allows users to upload these two folders directly. After uploading, the data are automatically segregated into individual unit-wise datasets along with the corresponding flare and controller datasets, thereby simplifying the subsequent monitoring and analysis. Snapshots of the developed GUI for unit-wise data handling and flare/controller data management are shown in Figures 6 and 7, respectively. The processed dataset is subsequently supplied for data preprocessing.

4.3 Data Preprocessing

The raw historian data obtained from the refinery contain measurements from multiple process units, flare system variables, controller data, and pressure safety valve (PSV) information. Prior to developing the monitoring models, the collected data undergo several preprocessing steps to improve data quality and ensure reliable model performance. The overall preprocessing workflow adopted in this work is illustrated in Figure 8. The preprocessing steps are summarized as follows:

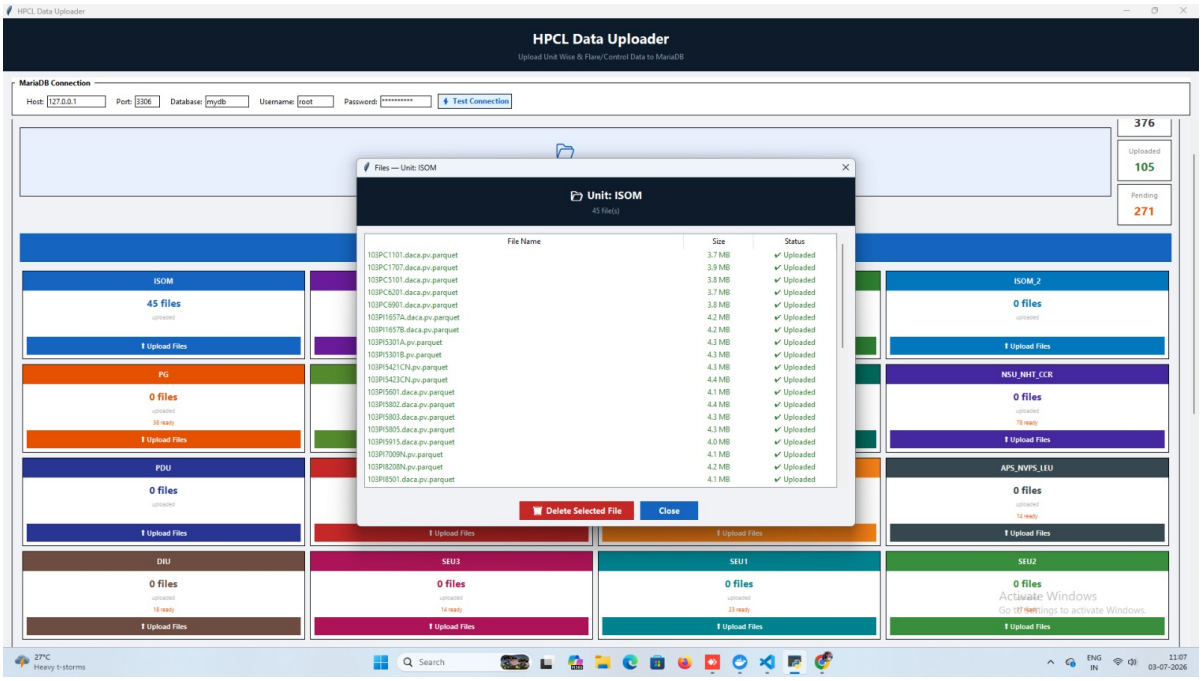


Figure 6: GUI for unit-wise data handling

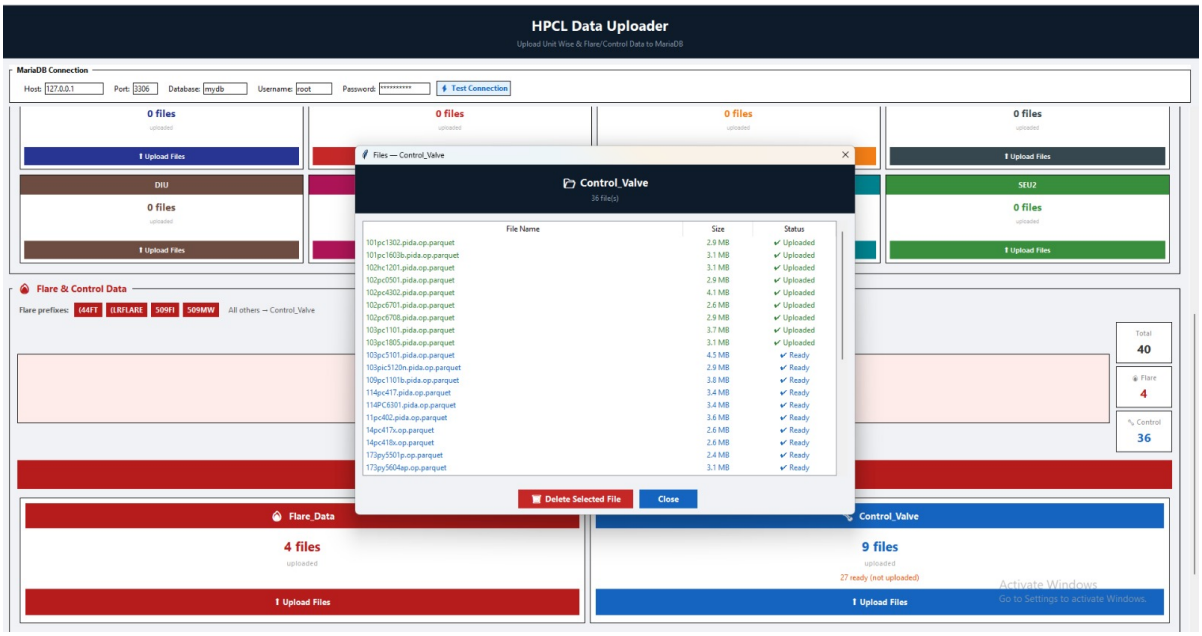


Figure 7: GUI for flare/controller data management

1. **Selection of Process Variables:** The required process variables are selected using the developed GUI. Unnecessary process tags, unavailable sensors, and variables not relevant to the monitoring objective are excluded from further analysis.

2. **Timestamp Synchronization:** Since all process measurements are recorded at one-minute intervals, the timestamps of all datasets are synchronized to generate a common time index. Duplicate timestamps, if present, are removed to ensure temporal consistency.

3. **Handling Missing Values:** The percentage of missing values is computed for every process variable. Variables containing a large proportion of missing measurements are

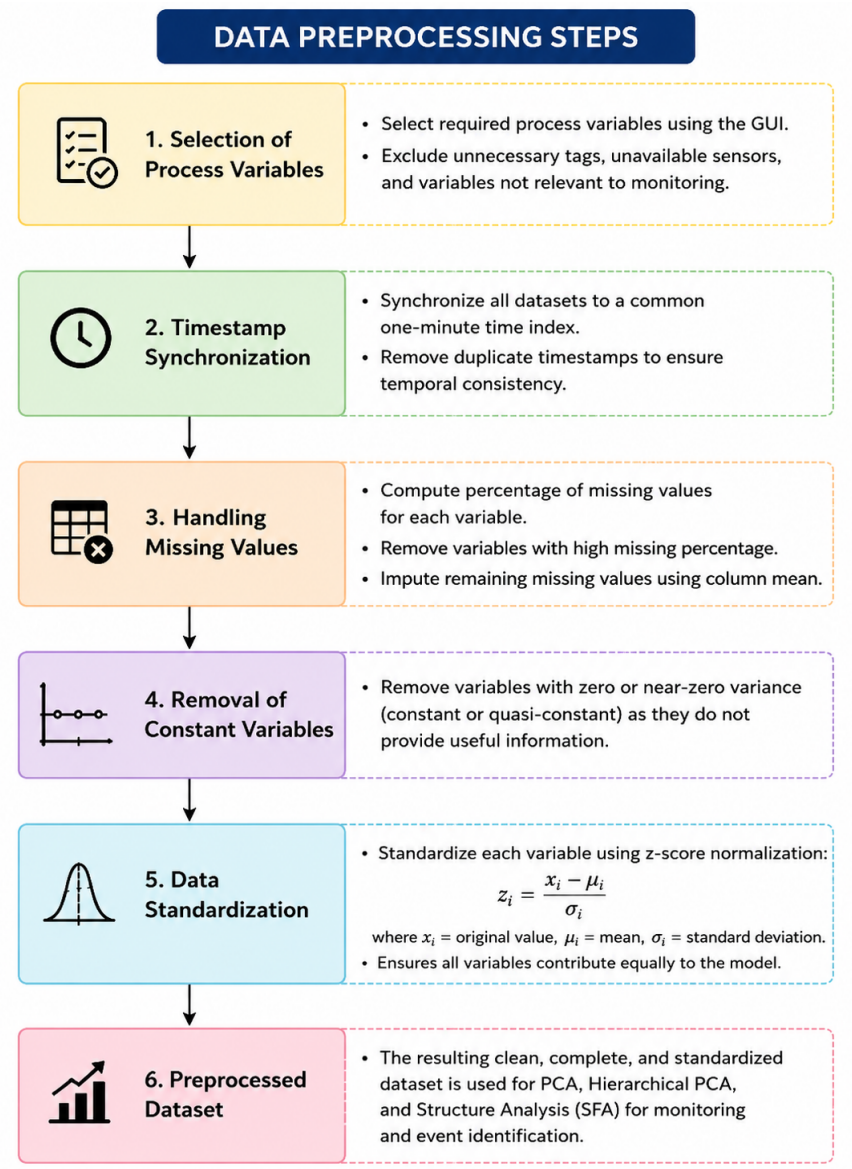


Figure 8: Flowchart of data preprocessing

discarded, whereas the remaining missing observations are imputed using the column mean to obtain a complete dataset suitable for statistical modelling.

4. **Removal of Constant Variables:** Variables exhibiting zero or near-zero variance throughout the monitoring period do not contribute useful information to multivariate analysis. Such variables are therefore removed from the dataset before model construction.

5. **Data Standardization:** Since the measured process variables possess different engineering units and operating ranges, each variable is standardized using z-score normalization prior to model development. For a process variable x_i, the standardized value is computed as

$$z_i = \frac{x_i - \mu_i}{\sigma_i}, \tag{46}$$

where μ_i and σ_i denote the mean and standard deviation of the ith process variable, re-

spectively.

Standardization ensures that all variables contribute equally during feature extraction and prevents variables with larger numerical magnitudes from dominating the monitoring model.

After completing the above preprocessing steps, the processed dataset is used for feature extraction using PCA, Hierarchical PCA, and Slow Feature Analysis (SFA), followed by statistical process monitoring and flare root cause identification.

4.4 Monitoring Strategy

The monitoring strategy constitutes the core of the proposed framework and is responsible for detecting abnormal process behaviour and identifying the variables contributing to flare events. After acquiring and preprocessing the process data, latent features are extracted from the selected process variables, and monitoring statistics are computed to identify abnormal operating conditions. Once an abnormality is detected, the flare event detection strategy described in the previous section is employed to evaluate whether the monitoring alarm successfully predicts an actual flare event. Subsequently, Reconstruction-Based Contribution (RBC) analysis is performed to determine the process variables responsible for the detected abnormality.

To investigate the effectiveness of different monitoring frameworks, four progressively advanced approaches are developed:

1. PCA-based Flare Root Cause Identification (Batch Mode)
2. PCA-based Flare Root Cause Identification (Moving Window)
3. Hierarchical PCA-based Flare Root Cause Identification (Moving Window)
4. Hierarchical PCA–SFA-based Flare Root Cause Identification (Moving Window)

4.5 Approach I: PCA-Based Flare Root Cause Identification (Batch Mode)

The first approach serves as the baseline methodology for flare root cause identification. A global PCA model is developed using the complete set of standardized process variables. The principal components are retained based on a predefined cumulative variance threshold, and the corresponding Hotelling's T^2 statistic is computed for each observation. Simultaneously, flare events are identified from the historian data using the flare event detection strategy described in Section 3.10. For each identified flare event, the monitoring statistic is examined within the predefined prediction horizon. A flare event is considered successfully detected if the T^2 statistic exceeds its control limit at least once during the 15-minute interval preceding the flare event. Reconstruction-Based Contribution (RBC) analysis is then performed only for the successfully detected flare events to identify the process variables responsible for the abnormal operating condition.

Although this approach provides a straightforward monitoring framework, it considers the refinery as a single process unit and does not explicitly account for the hierarchical structure of industrial plants or the non-stationary nature of refinery operation.

4.6 Approach II: PCA-Based Flare Root Cause Identification using Moving Window

In industrial process monitoring, process dynamics and operating conditions often change with time, making a static PCA model unsuitable for long-term monitoring. To address this issue, a moving window strategy is adopted in which the PCA model is continuously updated using the most recent observations. At each sampling instant, a fixed-size window containing the latest process data is considered. As a new sample becomes available, it is appended to the window while the oldest sample is discarded, ensuring that the window length remains constant. A new PCA model is then constructed using the updated window, allowing the principal components, eigenvalues, and monitoring statistics to adapt to gradual changes in the process. Consequently, the monitoring model reflects the current operating condition rather than historical behavior, improving the robustness of fault detection in non-stationary environments.

For flare root cause identification, the moving-window PCA model is used to monitor the evolution of process variables in real time. Hotelling's T^2 statistic is calculated for each incoming sample using the updated PCA model and compared with its corresponding confidence limit. When the T^2 statistic exceeds the control limit, the sample is identified as abnormal, indicating the occurrence of a flare-related event. Since the PCA model is updated at every window shift, the contribution of each process variable to the abnormality can be evaluated using the current loading structure. Variables exhibiting the highest contributions are considered the most likely sources of the flare event, enabling the identification of the root cause while minimizing the influence of outdated process information. This adaptive framework improves the reliability and accuracy of flare root cause identification under continuously varying operating conditions.

4.7 Approach III: Hierarchical PCA-Based Flare Root Cause Identification Using Moving Window

The proposed moving window hierarchical PCA framework is designed to continuously adapt the monitoring model to changing process operating conditions while simultaneously detecting flare events and identifying their root causes. Unlike the batch implementation, where a single PCA model is constructed using the entire dataset, the moving window approach updates both the local and global PCA models whenever a new process sample becomes available. Consequently, the monitoring model always represents the most recent process behavior, making it more suitable for non-stationary industrial processes.

The methodology begins by selecting a moving window of size W, where W represents the number of consecutive samples used for model development. The first window consists of the observations $\{x_1, x_2, \ldots, x_W\}$. For this initial window, the process data of each unit are first preprocessed by removing variables with zero variance, handling missing values, and standardizing each variable to have zero mean and unit variance. Subsequently, an independent PCA model is developed for each process unit, and the principal components satisfying the selected cumulative variance criterion are retained. The retained principal component scores from all units are then concatenated to form a plant-wide feature matrix, which is used to construct the global PCA model. The global Hotelling's T^2 statistic is computed for every sample within the first window, and the corresponding monitoring statistics are stored.

After processing the initial window, the monitoring framework operates sequentially using a moving window. When a new process sample x_t becomes available, the oldest observation in the current window is discarded, and the new sample is appended, thereby maintaining a constant window size of W. For example, if the current window contains $\{x_1, x_2, \ldots, x_W\}$,

the arrival of the sample x_{W+1} results in the updated window $\{x_2, x_3, \ldots, x_W, x_{W+1}\}$. Similarly, when the next sample x_{W+2} arrives, the sample x_2 is removed and the window becomes $\{x_3, x_4, \ldots, x_{W+1}, x_{W+2}\}$. This procedure is repeated until all process samples have been processed.

For every updated window, the unit-level PCA models are reconstructed using the latest process data, followed by reconstruction of the global PCA model using the concatenated principal component scores. Since the monitoring statistics for the previously processed observations have already been computed, Hotelling's T^2 statistic is evaluated only for the newly added sample. This significantly reduces the computational burden while preserving the adaptive nature of the monitoring framework.

Flare events are identified independently using the measured flare flow rate, Q_{flare}. A flare event is declared when the flare flow rate exceeds 1.5 for at least 15 consecutive samples. For each identified flare event, let k_s denote the starting instant of the flare. A prediction interval of $[k_s - 15,\ k_s]$, corresponding to a 15-minute prediction horizon prior to the flare occurrence, is examined using the previously computed Hotelling's T^2 statistics.

If at least one sample within the prediction interval satisfies

$$T^2(k) > T^2_{\text{lim}}, \tag{47}$$

the flare event is considered successfully detected. Hierarchical Reconstruction-Based Contribution (RBC) analysis is then performed to trace the detected abnormality from the global PCA model to the corresponding unit-level PCA models and finally to the original process variables. The variables are subsequently ranked according to their contribution values, thereby identifying the most probable root causes responsible for initiating the flare event. Conversely, if no sample exceeds Hotelling's T^2 control limit within the prediction interval, the flare event is classified as a missed detection.

The moving window implementation enables the monitoring model to continuously adapt to changes in process dynamics while avoiding redundant computation of monitoring statistics for previously analyzed observations. Consequently, it provides improved applicability for long-term online monitoring of industrial processes operating under slowly varying or drifting operating conditions.

4.8 Approach IV: Hierarchical PCA–SFA-Based Flare Root Cause Identification Using Moving Window

The moving window implementation is an adaptive monitoring strategy in which the PCA (or SFA) model is continuously updated using the most recent process data. Unlike the conventional batch implementation, where a single model is developed using the entire dataset, the moving window approach maintains a fixed window of size W that slides forward by one sample as new process measurements become available. This enables the monitoring model to adapt to slowly varying operating conditions while preserving a constant amount of training data.

Initially, the first W observations, $\{x_1, x_2, \ldots, x_W\}$, are selected to form the initial moving window. Using these samples, the PCA (or SFA) model is trained, and the monitoring statistics, such as Hotelling's T^2 and Reconstruction-Based Contribution (RBC), are computed for all samples within the first window. Since no previous monitoring results are available, the computed statistics for all W samples are stored.

Once the initial window has been processed, the window is shifted forward by one sample whenever a new observation becomes available. Suppose the current window consists of

$\{x_1, x_2, \ldots, x_W\}$. When the new sample x_{W+1} arrives, the oldest sample, x_1, is removed from the window and the newest sample, x_{W+1}, is appended, resulting in the updated window

$$\{x_2, x_3, \ldots, x_W, x_{W+1}\}. \tag{48}$$

Using this updated window, a new PCA (or SFA) model is trained to represent the latest operating condition of the process. Since the monitoring statistics for the retained samples $\{x_2, x_3, \ldots, x_W\}$ have already been computed in the previous iteration, only the newly added sample x_{W+1} is projected onto the updated PCA model to compute its Hotelling's T^2 statistic and RBC values. The computed statistics are then stored.

Similarly, when the next observation x_{W+2} becomes available, the oldest sample in the current window, x_2, is discarded and the new sample x_{W+2} is appended, producing the window

$$\{x_3, x_4, \ldots, x_{W+1}, x_{W+2}\}. \tag{49}$$

The PCA (or SFA) model is retrained using these updated samples, and the monitoring statistics are computed only for the newly added sample x_{W+2}. This procedure of removing the oldest observation, appending the newest observation, updating the monitoring model, and evaluating only the latest sample is repeated sequentially until the final process sample x_N has been processed.

The principal advantage of the moving window implementation is that the monitoring model continuously adapts to changes in process dynamics while avoiding unnecessary recomputation of monitoring statistics for samples that have already been evaluated. Consequently, the computational complexity of monitoring is significantly reduced, making the proposed framework suitable for online monitoring of large-scale industrial processes operating under slowly varying or drifting conditions.

5 Result and Discussion

This section presents the performance of the proposed hierarchical PCA-SFA-based framework for flare event detection and root cause identification using one year of refinery historian data. The results include data preprocessing, dimensionality reduction, flare event monitoring, validation using actual plant events, and interpretation of the identified root-cause variables. The effectiveness of the moving-window monitoring strategy is also discussed.

5.1 Data Preprocessing

The refinery historian database contained minute-wise measurements collected from 12 April 2025 to 21 May 2026, comprising 581,774 observations from 376 pressure and temperature process variables across multiple refinery units. For this study, data from 22 May 2025 to 21 May 2026 were considered, resulting in 523,084 observations. After preprocessing, 285 process variables were retained for subsequent analysis.

Figure 8 illustrates the data preprocessing workflow. Data preprocessing involved removing variables with a high percentage of missing values and eliminating variables with zero or near-zero variance. The remaining missing samples were imputed, and the process variables were synchronized based on their timestamps. Finally, the data were standardized to zero mean and unit variance before applying Principal Component Analysis (PCA), ensuring that variables with different engineering units contributed equally to the analysis.

Table 2: Number of process tags before and after preprocessing for each process unit.

Unit	Initial Tags	Final Tags
APS_NVPS_LEU	14	11
DAK	5	2
DIU	18	16
ISOM	45	41
ISOM3	14	14
ISOM4	3	2
LOUP	79	45
NSU-NHT-CCR	78	48
PDA	28	28
PG	38	34
SEU1	23	17
SEU2	17	14
SEU3	14	13
Total	**376**	**285**

5.2 Dimensionality Reduction using PCA-SFA

Table 3 4 present the dimensionality reduction achieved using the proposed hierarchical PCA-SFA framework. After preprocessing, 285 process variables from different refinery units were subjected to unit-wise PCA, where the number of variables was reduced to 114 principal components while retaining more than 95% of the variance for each unit. The retained principal components were subsequently combined and processed using global Slow Feature Analysis (SFA), resulting in 28 slow and 28 fast components. The slow components were used for process monitoring as they effectively capture the slowly varying dynamics associated with evolving process abnormalities. This hierarchical dimensionality reduction significantly decreases computational complexity while preserving the essential spatial and temporal information required for accurate flare event detection and root cause identification.

Table 3: Dimensionality reduction using unit-wise PCA. The number of principal components (PCs) retained for each refinery unit preserves more than 95% of the total variance.

Units	# Tags	# PCs (>95% variance)
APS_NVPS_LEU	11	7 (95.8%)
DAK	2	1 (100%)
DIU	16	11 (96.5%)
ISOM	41	15 (95.3%)
ISOM3	14	9 (96.3%)
ISOM4	2	1 (100%)
LOUP	45	8 (95.4%)
NSU-NHT-CCR	48	16 (95.1%)
PDA	28	10 (95.1%)
PG	34	12 (96.0%)
SEU1	17	3 (96.0%)
SEU2	14	11 (96.8%)
SEU3	13	10 (97.1%)
Total	**285**	**114**

Table 4: Summary of the dimensionality reduction achieved using the hierarchical PCA-SFA framework.

Unit-wise PCA	**Global SFA**	
Total PCs	**# Slow Components**	**# Fast Components**
114	28	28

5.3 Overall Monitoring Performance

The performance of the proposed moving-window PCA-SFA monitoring framework over the complete monitoring period is summarized in Figure 9. The figure presents the evolution of the slow T^2 statistic, fast T^2 statistic, and the recorded flare flow rate. The shaded regions indicate the actual flare episodes, while the dashed red line represents the corresponding control limit used for anomaly detection.

Table 5: Summary of flare event detection using the proposed PCA-SFA monitoring framework.

# Flare Events	**Slow T^2**	**Fast T^2**	**Both T^2**	**Total Detection**	**Detection Rate (%)**
2294	3	762	1377	2142	93.6

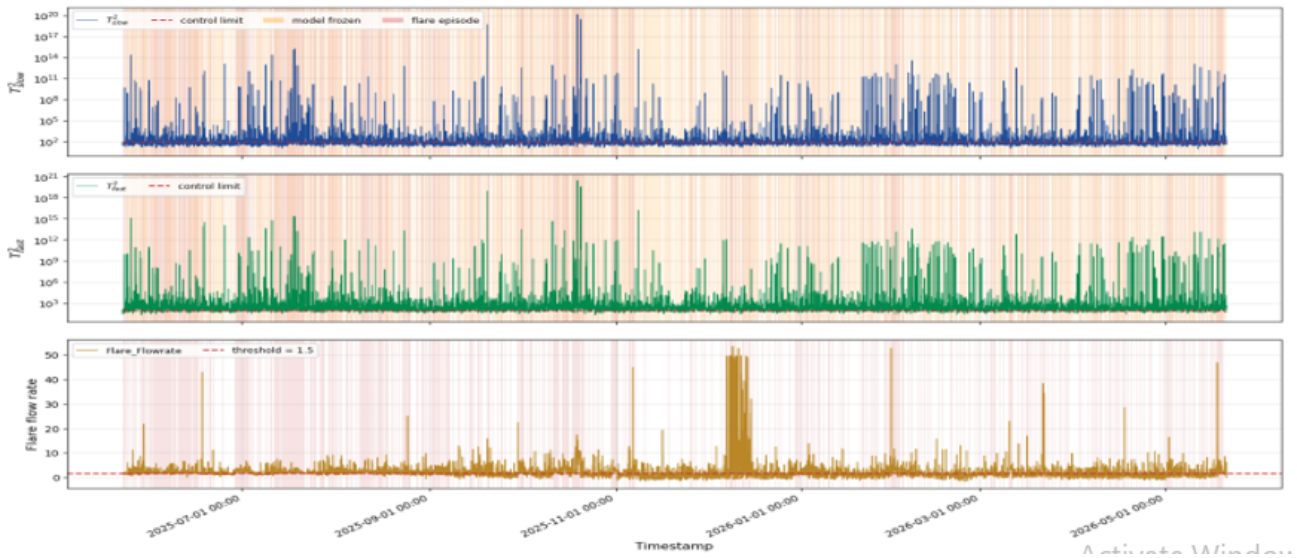


Figure 9: Monitoring results of the proposed PCA–SFA framework showing the slow and fast T^2 statistics and the corresponding flare flow rate. Shaded regions indicate flare events.

Table 5 summarizes the detection performance of the proposed monitoring framework. During the monitoring period, a total of 2294 flare events were recorded. Among these, 2142 events were successfully detected, resulting in an overall detection rate of 93.6%. Of the detected events, 1377 were identified simultaneously by both the slow and fast T^2 statistics, whereas 762 events were detected exclusively by the fast T^2 statistic. Only three events were detected solely by the slow T^2 statistic.

Figure 9 shows that the fast T^2 statistic exhibits frequent sharp peaks and responds rapidly to process disturbances, making it highly sensitive to abrupt changes in operating conditions. In contrast, the slow T^2 statistic varies more smoothly and captures the slowly evolving dynamics

of the process. During most flare episodes, simultaneous excursions in both monitoring statistics are observed, indicating that the abnormal operating conditions affect both the fast and slow process dynamics.

A comparison with the flare flow rate further confirms the effectiveness of the proposed monitoring framework. Most of the significant increases in flare flow coincide with elevated T^2 statistics, demonstrating a strong correlation between the monitoring indices and actual plant behaviour. Although a small number of flare events remain undetected, the overall detection rate of 93.6% indicates that the proposed hierarchical PCA-SFA framework provides reliable monitoring performance over long-term refinery operation. The complementary behaviour of the slow and fast T^2 statistics also suggests that utilizing both monitoring indices improves the robustness of flare event detection compared to relying on a single monitoring statistic.

5.4 Zoomed-in Analysis of Monitoring Statistics

To further evaluate the effectiveness of the proposed monitoring framework, a one-week segment of the monitoring results is presented in Figure 10. The zoomed-in view provides a clearer visualization of the relationship between the slow and fast T^2 statistics and the corresponding flare flow rate during individual flare episodes.



Figure 10: One-week monitoring results of the proposed method. The top and middle panels show the slow and fast T^2 statistics, and the bottom panel shows the flare flow rate. Shaded regions indicate flare events.

Figure 10 shows that both monitoring statistics remain close to their respective control limits during normal plant operation, indicating stable process conditions with only minor fluctuations. During flare episodes, however, significant excursions beyond the control limits are observed in both the slow and fast T^2 statistics, demonstrating the ability of the proposed framework to distinguish abnormal operating conditions from normal process behaviour.

The fast T^2 statistic exhibits sharp and high-amplitude peaks immediately following process disturbances, reflecting its sensitivity to rapid changes in the process variables. In contrast,

the slow T^2 statistic varies more gradually and captures the slowly evolving process dynamics associated with developing abnormalities. This complementary behaviour enables the monitoring framework to detect both abrupt disturbances and gradual process deviations.

A comparison with the flare flow rate further confirms the effectiveness of the proposed approach. Most periods of elevated flare flow correspond to simultaneous increases in the monitoring statistics, indicating a strong correlation between the statistical monitoring indices and the actual process condition. Moreover, noticeable increases in the T^2 statistics are observed before the flare flow reaches its maximum value during several events, suggesting that the proposed PCA-SFA framework is capable of providing an early indication of abnormal operating conditions. These results demonstrate the effectiveness of combining slow and fast features for reliable online monitoring and early flare event detection.

5.5 Summary of Final Outputs

The final output generated by the proposed hierarchical PCA-SFA framework is provided as a comma-separated values (CSV) file, enabling plant engineers to efficiently analyze detected flare events and their corresponding root causes. A snapshot of the generated output is shown in Figure 11. The CSV file consolidates the monitoring results into a structured format, making it suitable for both offline analysis and integration with existing refinery monitoring systems.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Event_ID	Event_Start	Event_Date	Event_Time	Event_End	Duration_min	Detected	Detected_By	Alarm_Onset	First_Hit_Timestamp	RBC_Source	RBC_Timestamp
2	1	2025-05-23 01:51:00	2025-05-23	01:51:00	2025-05-23 02:07:00	16	0					
3	2	2025-05-23 09:04:00	2025-05-23	09:04:00	2025-05-23 09:24:00	20	1	Both	2025-05-23 08:53:00	2025-05-23 08:53:00	fast	2025-05-23 08:53:00
4	3	2025-05-23 09:35:00	2025-05-23	09:35:00	2025-05-23 11:26:00	111	1	Both	2025-05-23 08:53:00	2025-05-23 09:20:00	fast	2025-05-23 08:53:00
5	4	2025-05-23 11:30:00	2025-05-23	11:30:00	2025-05-23 11:55:00	25	1	Both	2025-05-23 08:53:00	2025-05-23 11:15:00	fast	2025-05-23 08:53:00
6	5	2025-05-23 12:07:00	2025-05-23	12:07:00	2025-05-23 14:26:00	139	1	Both	2025-05-23 08:53:00	2025-05-23 11:52:00	fast	2025-05-23 08:53:00

	M	N	O	P	Q	R	S	T	U	V
1	Rank1_Variable	Rank1_RBC_Percent	Rank2_Variable	Rank2_RBC_Percent	Rank3_Variable	Rank3_RBC_Percent	Rank4_Variable	Rank4_RBC_Percent	Rank5_Variable	Rank5_RBC_Percent
2										
3	ISOM_PC5	7.79	ISOM_PC14	7.67	ISOM3_PC9	5.05	NSU_NHT_CCR_PC14	4.65	ISOM3_PC8	4.54
4	ISOM_PC5	7.79	ISOM_PC14	7.67	ISOM3_PC9	5.05	NSU_NHT_CCR_PC14	4.65	ISOM3_PC8	4.54
5	ISOM_PC5	7.79	ISOM_PC14	7.67	ISOM3_PC9	5.05	NSU_NHT_CCR_PC14	4.65	ISOM3_PC8	4.54
6	ISOM_PC5	7.79	ISOM_PC14	7.67	ISOM3_PC9	5.05	NSU_NHT_CCR_PC14	4.65	ISOM3_PC8	4.54

	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG
1	Rank5_RBC_Percent	TagRank1_Tag	TagRank1_Unit	TagRank1_RBC_Percent	TagRank2_Tag	TagRank2_Unit	TagRank2_RBC_Percent	TagRank3_Tag	TagRank3_Unit	TagRank3_RBC_Percent	TagRank4_Tag	TagRank4_Unit
2												
3	4.54	103PC5101.daca.pv	ISOM	11.59	103PI1657A.daca.pv	ISOM	6.87	42PC4275.PV	SEU3	4.64	14TI511.PV	ISOM3
4	4.54	103PC5101.daca.pv	ISOM	11.59	103PI1657A.daca.pv	ISOM	6.87	42PC4275.PV	SEU3	4.64	14TI511.PV	ISOM3
5	4.54	103PC5101.daca.pv	ISOM	11.59	103PI1657A.daca.pv	ISOM	6.87	42PC4275.PV	SEU3	4.64	14TI511.PV	ISOM3
6	4.54	103PC5101.daca.pv	ISOM	11.59	103PI1657A.daca.pv	ISOM	6.87	42PC4275.PV	SEU3	4.64	14TI511.PV	ISOM3

Figure 11: Snapshot of the final CSV output generated by the proposed monitoring framework. The file contains event information, alarm details, and ranked root-cause variables associated with each detected flare event.

As shown in Figure 11, the first section of the CSV file provides event-level information, including the event ID, event start and end times, event duration, detection status, detection source (slow T^2, fast T^2, or both), alarm onset time, first hit timestamp, and the corresponding root-cause identification (RBC) time. These details enable operators to quickly assess when an abnormal event occurred and how it was detected.

The second section reports the ranked principal components contributing to each flare event along with their corresponding contribution percentages. This information identifies the latent

variables that are primarily responsible for the abnormal process behaviour.

Finally, the identified principal components are mapped back to the original process variables and refinery units. The output lists the top contributing process tags together with their respective units and contribution percentages, thereby providing operators with a clear indication of the equipment or process section most strongly associated with the flare event. This ranked information facilitates rapid fault diagnosis by directing plant personnel toward the most influential variables instead of requiring manual examination of hundreds of process tags.

Overall, the generated CSV file transforms the statistical monitoring results into an interpretable and operator-friendly format, providing comprehensive information on event detection, alarm generation, and ranked root-cause variables for each detected flare event. This significantly reduces the time required for fault investigation and supports informed decision-making during refinery operation.

5.6 Algorithm validation

5.6.1 Scenario 1

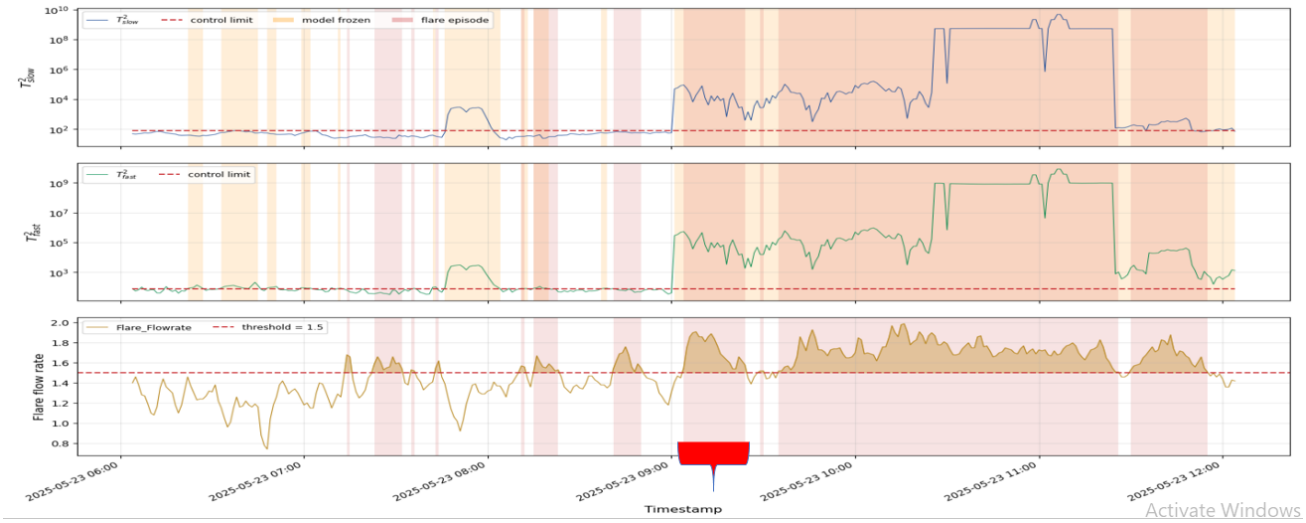


Figure 12: Monitoring results for the flare event on 23 May 2026. The top and middle plots show the slow and fast T^2 statistics, and the bottom plot shows the flare flow rate. The shaded region indicates the flare event.

Figure 12 illustrates the performance of the proposed PCA-based flare root cause identification using a moving window for a flare event that occurred between 9:04 AM and 9:24 AM on 23 May 2026. The upper and middle plots show Hotelling's T^2 statistics computed using the slow and fast PCA models, respectively, while the bottom plot presents the corresponding flare flow rate. The red dashed lines in the first two plots denote the statistical control limits, whereas the red dashed line in the flare flow plot represents the flare threshold of 1.5. It can be observed that both T^2 statistics remain below their respective control limits during normal operation and exceed the threshold at approximately 8:53 AM, indicating the onset of an abnormal process condition.

Following the detection of the abnormal condition, contribution analysis is performed using the updated PCA model to identify the source of the fault. The analysis reveals that ISOM_PC5 and ISOM_PC14 are the dominant principal components contributing to the abnormality. Further investigation of the corresponding loading vectors identifies the process variables

103PC5101.daca.pv and 103PI1657A.daca.pv as the most significant contributors, indicating that these tags are the most probable root causes of the flare event. These results demonstrate that the proposed moving-window PCA approach not only detects flare events well before their occurrence but also accurately identifies the associated root-cause variables, thereby providing valuable information for timely operator intervention and effective process troubleshooting.

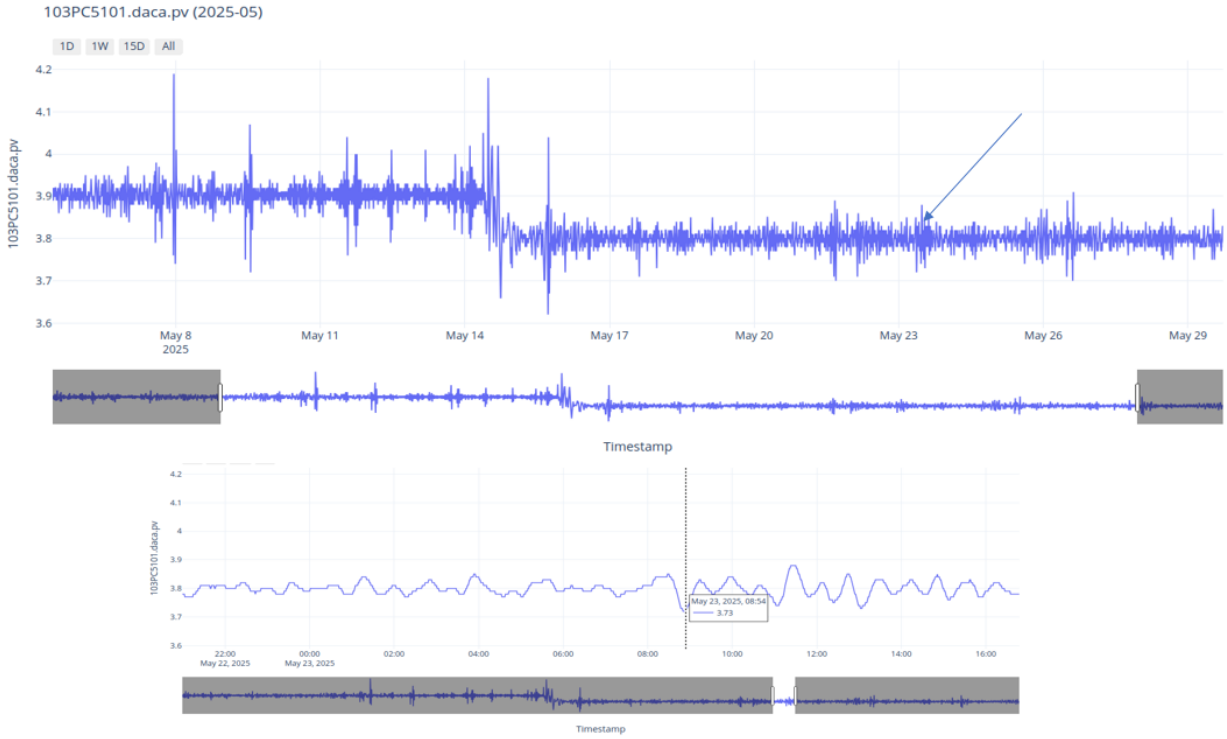


Figure 13: Trend of 103PC5101.daca.pv around the flare event on 23 May 2025, showing the abrupt dip at approximately 08:54 AM.

Figure 13 shows the temporal evolution of the process variable 103PC5101.daca.pv, which was identified as one of the highest contributing variables during the flare root cause analysis. The upper plot presents the complete trend for May 2025, while the lower plot provides a zoomed view of the period surrounding the flare event on 23 May 2025. It can be observed that the process variable remains relatively stable under normal operating conditions with only small fluctuations. However, at approximately 08:54 AM on 23 May 2025, the tag exhibits a distinct and abrupt decrease from its nominal operating value. This deviation occurs shortly before the flare event and closely coincides with the alarm generated by the proposed PCA-based monitoring algorithm. The sudden dip indicates a significant process disturbance that is captured by the moving-window PCA model, resulting in a high contribution of this variable to Hotelling's T^2 statistic. Consequently, 103PC5101.daca.pv is identified as one of the most probable root-cause variables associated with the flare event. The close temporal correspondence between the abnormal change in this process variable and the onset of the flare further validates the effectiveness of the proposed contribution analysis in localizing the source of the abnormal operating condition.

Figure 14 presents the trend of the process variable 103PI1657A.daca.pv, which was identified as one of the highest contributing variables during the PCA-based root cause analysis. The process variable remains relatively stable throughout the monitoring period, indicating normal operation with only minor fluctuations. However, just before 09:06 AM on 23 May 2025, a

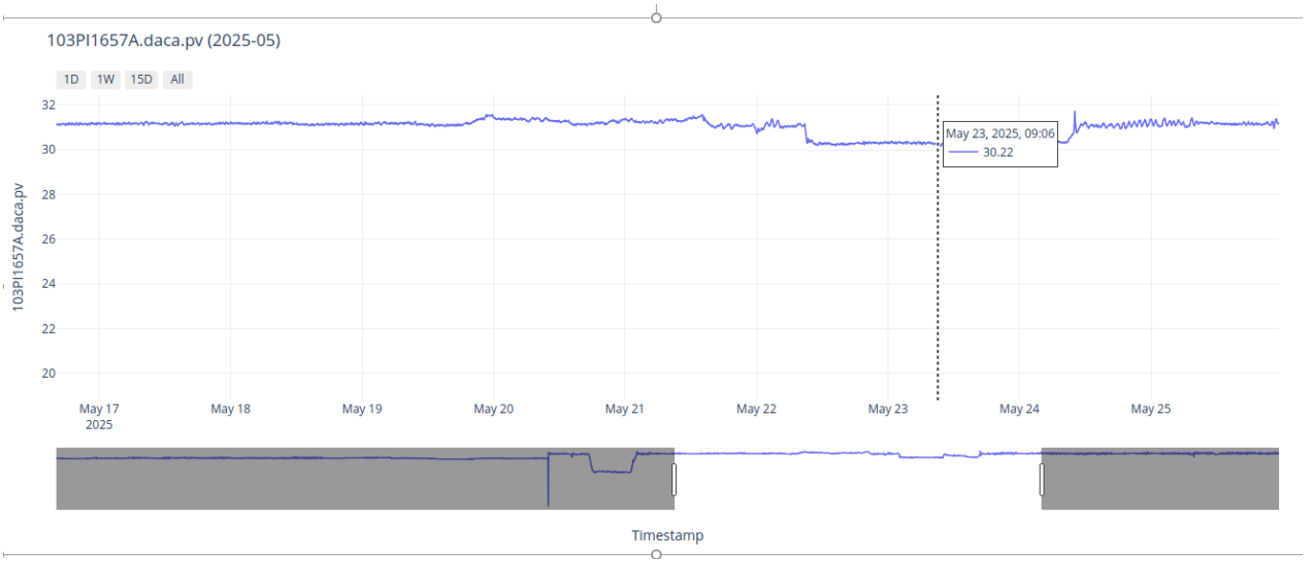


Figure 14: Trend of `103PI1657A.daca.pv`, highlighting the deviation at approximately 09:06 AM on 23 May 2025.

noticeable decrease in the signal is observed immediately before the flare event. Although this deviation may represent a measurement artefact or a transient process disturbance, its occurrence coincides with the period during which the monitoring statistics exceed their control limits. Consequently, the moving-window PCA model assigns a high contribution to this variable, suggesting that it is strongly associated with the abnormal operating condition. Even if the observed dip is not the direct physical cause of the flare, such abrupt changes significantly influence the principal component scores and Hotelling's T^2 statistic, thereby making `103PI1657A.daca.pv` an important indicator for flare detection. The close temporal relationship between this deviation and the detected abnormality demonstrates the ability of the proposed contribution analysis to identify process variables that are most strongly associated with the onset of a flare event.

5.6.2 Scenario 2

Figure 15 presents the second case study for the proposed PCA-based flare root cause identification approach. The upper plot shows the flare flow rate, where a flare event is observed from 10:34 AM on 12 July 2025. The increase in flare flow, highlighted by the red arrow, indicates the onset of the abnormal operating condition. Following the detection of this event, the contribution analysis was performed using the moving-window PCA model to identify the process variable most strongly associated with the abnormality.

The contribution analysis identified `71TI1703.PV` as the dominant process variable responsible for the abnormal condition. As shown in the lower plot of Figure 15, the process variable remains relatively stable before the flare event but exhibits a significant downward trend immediately after the flare begins. The signal continues to decrease over the following hours before gradually recovering towards its normal operating range. This sustained deviation from its nominal operating condition results in a large contribution to Hotelling's T^2 statistic, causing `71TI1703.PV` to be ranked as the most significant contributor by the moving-window PCA model. The close temporal correlation between the deviation of this process variable and the flare event suggests that it is strongly associated with the abnormal process condition that led to the flare. These results demonstrate that the proposed moving-window PCA framework not only detects flare events but also effectively identifies the process variables that contribute most



Figure 15: Case 2: Root cause analysis of the flare event on 12 July 2025, showing the flare flow rate and the identified process variable `71TI1703.PV`.

significantly to the observed abnormality, thereby assisting plant operators in diagnosing the underlying cause of the flare. The signal decreases significantly over the subsequent hours before gradually recovering to its normal operating range. This sustained deviation from its nominal operating condition results in a large contribution to Hotelling's T^2 statistic, causing the variable to be ranked as the most significant contributor by the PCA model. The strong temporal correlation between the change in `71TI1703.PV` and the flare event suggests that this process variable is closely associated with the process disturbance that triggered the flare. These results further demonstrate that the proposed moving-window PCA framework is capable of not only providing early detection of flare events but also effectively identifying the process variables most likely to be responsible for the abnormal operation, thereby assisting plant operators in diagnosing the underlying cause of the flare.

5.7 Comparison of Monitoring Algorithms

To evaluate the effectiveness of the proposed monitoring framework, five different algorithms were implemented and tested using the same refinery dataset. Table 6 summarizes their flare event detection performance.

Table 6: Comparison of flare event detection performance for different monitoring algorithms.

Algorithm	Total Flare Events	# Detected	Detection Rate (%)
I	2294	262	11.4
II	2294	985	42.9
III (2 PCs/Unit + 1 Global PC)	2294	920	40.1
IV (2 PCs/Unit + SFA)	2294	1395	60.3
V (>95% variance/unit)	**2294**	**2142**	**93.6**

Table 6 compares the flare detection performance of the different monitoring strategies investigated in this work. Algorithm I, which employs a batch-mode PCA-based monitoring approach,

detected only 262 out of 2294 flare events, corresponding to a detection rate of 11.4%. Updating the PCA model using a moving-window strategy in Algorithm II significantly improved the detection rate to 42.9%, highlighting the importance of adapting the monitoring model to continuously changing process conditions.

Algorithm III introduced a hierarchical PCA-based framework, where each refinery unit was represented using two principal components followed by a global principal component analysis. Although this approach improved the interpretability of the monitoring framework, its detection rate (40.1%) remained comparable to that of Algorithm II. In Algorithm IV, Slow Feature Analysis (SFA) was incorporated into the hierarchical PCA framework while retaining two principal components per unit. The inclusion of temporal process information improved the detection rate to 60.3%, demonstrating the advantage of combining spatial and temporal information for flare event monitoring.

Finally, instead of restricting each unit to two principal components, the proposed framework retained all principal components explaining more than 95% of the variance from each refinery unit before applying Global SFA. This resulted in a substantial improvement in monitoring performance, with 2142 out of 2294 flare events successfully detected, corresponding to an overall detection rate of 93.6%. The significant improvement demonstrates that preserving sufficient process information prior to SFA is essential for accurately capturing abnormal operating conditions. Overall, the proposed hierarchical PCA-SFA framework with variance-based principal component selection achieved the highest detection accuracy among all the investigated monitoring strategies, making it the most suitable approach for reliable online flare event detection.

6 Conclusion

- Four monitoring algorithms were developed and evaluated for early flare event detection using refinery process data containing 2,461 flare events.
- Among the proposed methods, Algorithm V achieved the highest detection performance by identifying 2142 flare events, corresponding to a detection rate of 93.6%, outperforming Algorithms I (11.4%), II (42.9%), and III (60.3%),IV(60.3%) as presented in Table 6.
- The improved performance of Algorithm IV demonstrates the advantage of combining slow- and fast-feature monitoring statistics for capturing diverse process dynamics associated with flare events.
- Reconstruction-Based Contribution (RBC) analysis successfully identified the dominant process variables responsible for the detected flare events, thereby facilitating effective root-cause identification.
- The proposed framework provides an interpretable and practical approach for refinery flare event detection and diagnosis, enabling timely operator intervention to improve process safety and operational reliability.

7 Future Directions

- Algorithm 4 needs further verification through additional sanity checks.

- Explore **Dynamic Principal Component Analysis (DPCA)** to incorporate temporal process dynamics and improve the early detection of evolving flare events.
- Investigate **supervised machine learning approaches**, such as XGBoost, Random Forest, Support Vector Machines (SVM), and deep learning models, to leverage historical labelled flare event data for improved detection accuracy and prediction performance.
- Integrate **Causal AI** techniques, including the Peter and Clark Momentary Conditional Independence (PCMCI) algorithm, to uncover causal relationships among process variables, thereby enhancing root-cause identification and supporting explainable decision-making.

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